

## Response to the reviewers

Manuscript: *Structure-Aware Row Scaling for Block-Structured Mixed-Integer Programs*  
(submitted under the title *Generator-Aware Positive Row Scaling for Block-Structured Engineering Mixed-Integer Programs*)

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Manuscript ID: [TO FILL]

Dear Editor and Reviewers,

Thank you for the careful and constructive evaluation. We have revised the manuscript following every required (R) and suggested (S) point. The revision adds three new experimental controls run on the same cluster and instance pools as the original study (a solver-internal-scaling baseline, a hyperparameter sensitivity sweep, and an operational stage ablation, now covering all three instance classes), plus a fourth control that the ablation itself motivated (a role-blind “flat” scaling control), a full original-scale equivalence audit of the four operational campaigns, and tempered claims wherever the new evidence demanded it. All new code paths are additive and opt-in: without the new flags the binaries are byte-identical to the version that produced the published results (golden-file regression 23/23), so the original tables remain valid. A marked manuscript (`latexdiff`) against the submitted version accompanies this letter. The manuscript was retitled to *Structure-Aware Row Scaling for Block-Structured Mixed-Integer Programs*; the reproducibility repository was renamed accordingly (<https://github.com/MathiasRodriguezCastro/structure-aware-row-scaling>, the old URL redirects) and its citation metadata updated.

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## Required revisions

### R1 — Solver-internal-scaling baseline

*Is the reported gain just a generic scaling effect that the solver could have produced itself with its own scaling controls? [replace with verbatim comment]*

**Response.** We agree this is the decisive control and have run it in full: the unscaled model under every admissible internal-scaling setting of each solver (Gurobi `ScaleFlag` 0–3; CPLEX `Read::Scale` –1/0/1), on the same instances, for all three classes and both tolerances, compared against the structure-aware variants under default internal scaling. Three findings. (i) Internal scaling is not a straw man: disabling it is catastrophic (e.g. Simple Gurobi `sgm` 3.9 → 42 s). (ii) On the classes where the external rule is effective, no tuned internal setting reproduces its runtime: the best internal Gurobi flag returns to the default level while the structure-aware rule is 1.8× faster (Simple), and no internal mode comes close on SG-Ter-Mer. (iii) We also report honestly where this control does *not* favor the method: on the Full class at the strictest tolerance the retuned internal modes overtake the external rule, and on Full under CPLEX the “no scaling” setting appears dramatically faster but produces claimed-optimal objectives that disagree with the default pipeline beyond the stopping tolerance on a third of the instances—we flag it as numerically unreliable, which itself illustrates the paper’s point that scaling choices can invalidate certificates.

**Changes in the manuscript.** New design paragraph in Section 5.4 (additional controls); results in Section 6.7 with the extended Table 17 (three classes at 1%; the 0.1% sweep summarized in text); Discussion updated to scope the claim accordingly. Canonical CSVs under `results-revision/r1-internal-scaling/` in the repository.

## R2 — Conditioning vs. runtime inconsistency

*SA-Aug is better conditioned but SA-Mat is often faster; conditioning does not line up with runtime. [replace with verbatim comment]*

**Response.** We now state the variant-level inversion explicitly and re-frame conditioning as at most a *partial mediator* of the runtime effect throughout (abstract, results lead-in, Discussion). The revision strengthens this further with two additions: the correlation table now reports the per-variant decomposition—the pooled negative conditioning–speedup association is carried by the Ruiz baselines ( $\rho_S \approx -0.33/-0.36$ ), whereas for the structure-aware variants it is essentially null—and the stage ablation (R5) shows that the stage that most improves the coupling diagnostics is not the stage that drives the runtime. Precedent for both observations is now cited (Elble & Sahinidis 2012; Miltenberger, Ralphs & Steffy 2018).

**Changes in the manuscript.** Section 5.6 (statistical analysis), Table 9 notes and surrounding text; Discussion paragraph “A mechanistic caveat. . .”; new references.

## R3 — Hyperparameter sensitivity

*The constants  $u = 2$ ,  $K = 4$  and the coefficient window were fixed without justification. [replace with verbatim comment]*

**Response.** The constants are now exposed as flags and swept on the Simple class under Gurobi:  $u \in \{1.5, 2, 4\}$ ,  $\text{window} \in \{[10^{-6}, 10^6], [10^{-9}, 10^9], [10^{-12}, 10^{12}]\}$ ,  $K \in \{2, 4, 8\}$ . The structure-aware rule is insensitive ( $\text{sgm} \in [2.01, 2.12]$  s across the whole grid against a base of 3.73 s). We note transparently that  $K$  is a parameter of the Ruiz baselines only; varying it leaves the structure-aware rule unchanged, as expected.

**Changes in the manuscript.** Section 5.4 (design) and Section 6.9 with Table 19; canonical CSVs under results-revision/r3-hyperparam/.

## R4 — Generality and the row-only premise

*The generality claims should be rescoped; the row-only choice is a premise, not a finding. [replace with verbatim comment]*

**Response.** The abstract now states that all runtime evidence is single-platform and the method is demonstrated on one hydrothermal testbed; a third limitation makes the row-only premise explicit and identifies structured *column* scaling as an untested companion lever; the generalization to other domains is qualified by the shared structural premise (sparse global coupling over local blocks). The title was kept accurate to the method, with the scope made explicit in the text.

**Changes in the manuscript.** Abstract; Section 8 (limitations), third limitation; Conclusion.

## R5 — Operational stage ablation

*Local-only and coupling-only variants were never run on the real hydrothermal classes.  
[replace with verbatim comment]*

**Response.** Both single-stage ablations now run on the real classes—including the Full class—at both tolerances under Gurobi. The attribution is unambiguous and consistent: the local stage alone recovers essentially the entire runtime benefit (Simple 3.73 → 2.01 s vs. 2.02 for the full rule; Full 82.6 → 70.9 vs. 71.4; SG-Ter-Mer 14.4 → 8.6 vs. 8.6), while the coupling stage alone is inert. We report this even though it qualifies our own architecture, and we drew the natural consequence: since the local kernel is computable from the flat matrix, we added a further *role-blind flat control* (the same per-row kernel and safeguards applied to every row with no metadata at all) to isolate whether the row-role metadata is the active ingredient of the runtime effect. [TO FILL when the campaign completes: flat control outcome and the corresponding scoping of the performance claim.]

**Changes in the manuscript.** Section 5.4 (design), Section 6.8 with Table 18 (three classes); Conclusion updated (the previously listed “future work” ablation is now in the paper); flat control code and campaign in the repository (-plano).

## R6 — Related-work positioning

*The related work should engage the classical equilibration literature and state precisely what is new. [replace with verbatim comment]*

**Response.** The related work now engages geometric-/arithmetic-mean scaling (Tomlin), Curtis–Reid, Sinkhorn–Knopp, Ruiz, and solver presolve (Achterberg et al.), and states precisely what is new: not the kernel (the local factor is a geometric mean of per-row extremes) but the metadata-driven *source* of the scales and the staged local–block–coupling *ordering*. The revision adds Elble & Sahinidis (2012) as the empirical precedent that generic scaling payoffs are irregular, Miltenberger et al. (2018) on conditioning along the branch-and-cut tree, and Bergner et al. (2015) for automatic structure detection as the fallback when the generator is not owned; the absolute claim about modeling languages exporting flat matrices is now qualified (block-structured extensions exist, but they feed decomposition algorithms rather than numerical preprocessing—exactly the gap addressed).

**Changes in the manuscript.** Section 2.3–Section 2.4; Section 7 (deployability); references.

## Suggested revisions

### S1 — Deployability

**Response.** New Discussion subsection *Deployability and downstream effects*: metadata availability in Pyomo/JuMP/GAMS/AMPL, instrumentation cost for owned generators, the detection fallback (now cited), and a deliberately narrow value proposition.

## S2 — Dual variables

**Response.** The same subsection now states that positive row scaling rescales constraint duals by the inverse row factors, that the factors are stored, and that operators must descale shadow prices (marginal costs, water values) before use.

## S3 — Proposition–proxy bridge

**Response.** The propositions are explicitly downgraded to motivation, not guarantees; the synthetic suite is identified as the empirical bridge, and the revision adds the per-block audit trail via the controlled synthetic mechanism check.

## S4 — Abstract effect-size language

**Response.** The abstract describes the Gurobi benefit as concentrated timeout reduction rather than a broad per-instance speedup; the single-instance PAR10 caveat of SG-Ter-Mer is now stated in the notes of *both* tolerance tables.

## S5 — Partial correlations and missingness

**Response.** The statistical section reports the size-controlled partial Spearman ( $\rho \approx -0.25$ ,  $n = 1185$ , essentially equal to the raw value), the balanced pools, the 1–3% per-run error rate, and the matched-pairs policy; the revision adds the per-variant decomposition of the pooled correlation (see R2).

## Editor’s comments

### Length (W4)

*The manuscript should be shortened by roughly 15–20%. [replace with verbatim comment]*

**Response.** [T0 FILL after the P2 editing pass: current status is that the required controls added substantiating material; the dedicated trim pass consolidates the local-block-coupling framing and the three-claims statements to single occurrences, merges the two claim-summary tables, compresses the post-table narration paragraphs, and moves the redundant 0.1% Gurobi figures to Supporting Information.]

## New material added on our own initiative

- *Original-scale equivalence audit* (Section 6.1, Table 6): the descaling verification promised by the design is now reported for all four operational campaigns, with verified explanations of every exceedance; notably, the audit shows the strict checker fails the *unscaled base* at the same rate as the scaled variants where exceedances occur, and it catches wrong-optimal certificates in both directions.
- CPLEX tree-level KappaStats, so the numerical claim is now tabulated under both solvers.
- A run-to-run performance-variability paragraph in the limitations, using the repeated base campaigns as an empirical noise bound.

We believe the revision addresses every point and, where the new controls qualified our own claims, we have reported those qualifications explicitly. We thank the panel again for a review that materially improved the paper.

Sincerely,

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